

Quantum Field Theory — Lecture 3

IMAPP lecture notes

Spring 2026

1 Fields and the action

A classical field assigns a quantity $\phi(x)$ to every point of a D -dimensional spacetime. The dynamics are encoded in a local Lagrangian density,

$$\mathcal{L} = \mathcal{L}(\phi, \partial_\mu \phi; x),$$

and the action is

$$S[\phi] = \int_{\Omega} d^D x \mathcal{L}(\phi, \partial_\mu \phi; x).$$

Locality means that the density depends on the field and its derivatives at one spacetime point.

2 Euler-Lagrange equations

Vary the field, $\phi \rightarrow \phi + \delta\phi$, and require the action to be stationary. To first order,

$$\delta S = \int d^D x \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu \delta\phi \right).$$

Integrating the second term by parts gives a bulk term and a boundary term:

$$\delta S = \int d^D x \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \right] \delta\phi + \text{boundary term}.$$

For variations that vanish on the boundary, the Euler-Lagrange equation is

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) = 0.}$$

3 Functional derivatives

A functional maps a function to a number, $F : \{\phi\} \rightarrow \mathbb{R}$. Its functional derivative is defined by

$$\frac{\delta F[\phi]}{\delta \phi(x)} = \lim_{\varepsilon \rightarrow 0} \frac{F[\phi + \varepsilon \delta_x] - F[\phi]}{\varepsilon},$$

where the Dirac delta plays the role of a unit variation. In particular,

$$\frac{\delta \phi(y)}{\delta \phi(x)} = \delta^{(D)}(x - y).$$

For a quadratic functional,

$$F[\phi] = \int d^D x \, d^D y \, \phi(x) G(x, y) \phi(y),$$

the derivative receives contributions from both occurrences of ϕ .

4 Hamiltonian formalism

For a discrete coordinate q_i , the conjugate momentum is $p_i = \partial L / \partial \dot{q}_i$ and $H = \sum_i p_i \dot{q}_i - L$. In the continuum, define the momentum density

$$\pi(x) = \frac{\partial \mathcal{L}}{\partial (\partial_t \phi(x))},$$

and the Hamiltonian density

$$\mathcal{H} = \pi \dot{\phi} - \mathcal{L}, \quad H = \int d^{D-1}x \mathcal{H}.$$

The Hamiltonian singles out time, whereas the Lagrangian formalism treats time and space in a more symmetric way.

5 Symmetries

Transformations may be continuous or discrete. Continuous transformations can be infinitesimal and may act on coordinates, fields, or both. A transformation is a symmetry when it leaves the action invariant:

$$S[\phi'] = S[\phi].$$

The invariance can be exact or can differ by a total divergence. Under the usual boundary assumptions, a total divergence contributes only a boundary term and does not change the equations of motion. This observation leads to Noether's theorem: every continuous differentiable symmetry is associated with a conserved current.

Summary

The action principle supplies the field equations, functional derivatives generalize ordinary differentiation to field space, and the Hamiltonian is obtained by a Legendre transform. Continuous action symmetries are the starting point for conserved quantities.